2026
FIFA WORLD CUP

DBD2781Project – GROUP 13



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Introduction and Problem Statement

Introduction

The FIFA World Cup 2026 database system is designed to manage and organize all tournament-related data efficiently. The system stores and processes information about countries, cities, stadiums, teams, players, matches, and ticket sales. By centralizing this data into a structured relational database, the system ensures consistency, accuracy, and ease of access.

Problem Statement

Managing World Cup data manually or using unstructured systems can lead to redundancy, inconsistencies, and data errors. For example, duplicate match records, incorrect score entries, and mismatched team data can compromise the reliability of the system. Additionally, handling large volumes of ticket sales and match scheduling requires a scalable and secure solution.

This database addresses these challenges by:

- Eliminating redundancy through normalization
- Enforcing data integrity using constraints and triggers
- Supporting efficient data retrieval through structured queries
- Providing secure access control through user roles

Business Rules

The FIFA World Cup 2026 database is designed to manage information related to teams, players, matches, stadiums, and ticket bookings. The following business rules define how the system operates:

1. Each country can have multiple cities, but each city belongs to only one country.
2. Each city can contain multiple stadiums, but each stadium is located in only one city.
3. Each team represents a specific country and is assigned to a group (A–F).
4. Each team can have multiple players, but each player belongs to only one team.
5. Each match is played between two teams (home and away) and takes place in one stadium.
6. Each match has a date, time, stage, and score (home and away goals).
7. Each stadium can host multiple matches.
8. Fans can purchase tickets for matches, and each ticket is linked to one specific match.
9. A match can have multiple tickets sold to different fans.
10. Scores for matches cannot be negative (enforced using a trigger).
11. Duplicate matches (same teams on the same date) are not allowed.

These rules ensure data integrity and reflect real-world tournament operations.

Entity Relationship Diagram (ERD) Description

The database consists of the following main entities:

- Countries
- Cities
- Stadiums
- Teams
- Players
- Matches
- Tickets

Relationships:

Countries → Cities

One-to-Many (1:M)

A country can have many cities.

Cities → Stadiums

One-to-Many (1:M)

A city can have multiple stadiums.

Countries → Teams

One-to-Many (1:M)

Each team belongs to one country.

Teams → Players

One-to-Many (1:M)

Each team has multiple players.

Teams to Matches

One-to-Many (1:M)

Teams participate in matches as home or away teams.

Stadiums to Matches

One-to-Many (1:M)

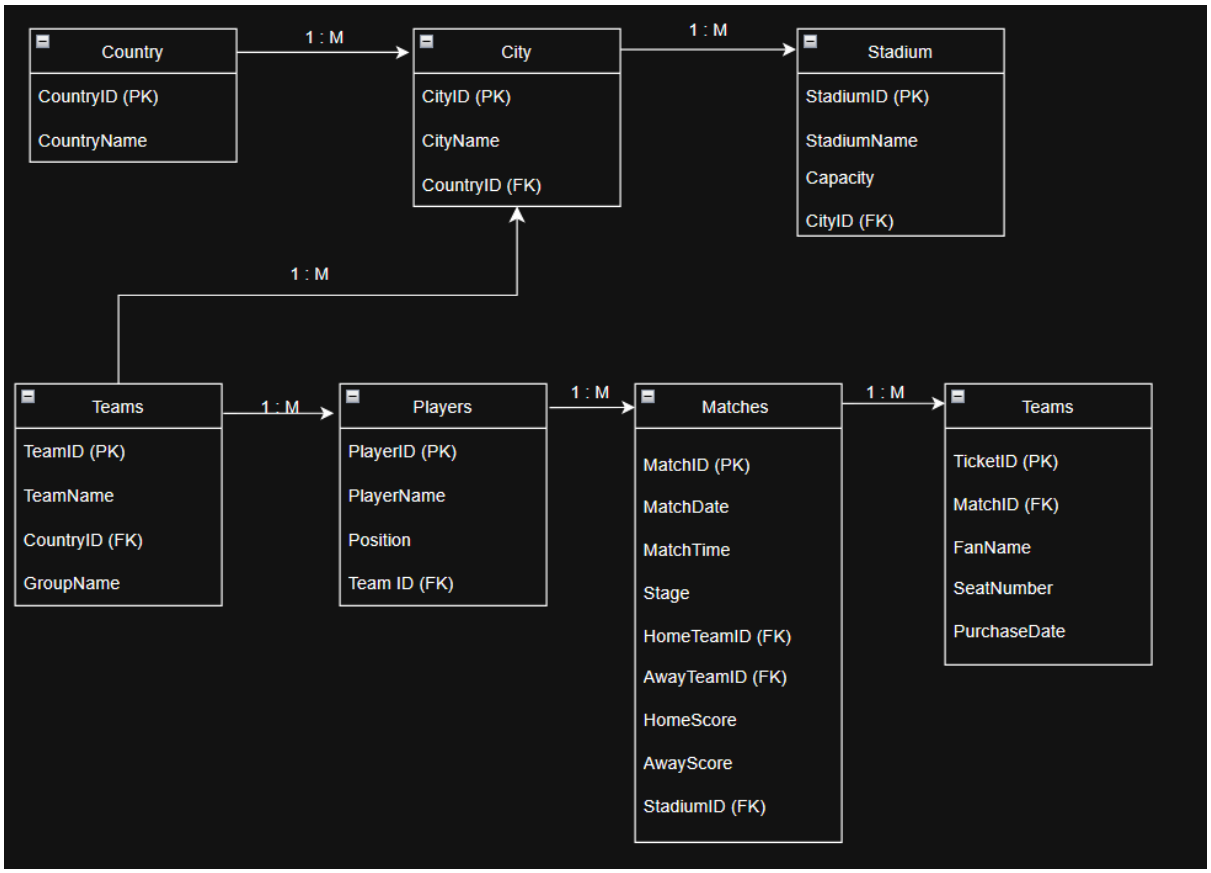
Each match is played in one stadium.

Matches to Tickets

One-to-Many (1:M)

A match can have many tickets sold.

ERD Representation



Normalization (Up to Third Normal Form 3NF)

The database has been normalized to ensure efficiency and eliminate redundancy.

First Normal Form (1NF)

All tables have primary keys.

No repeating groups or multi-valued attributes exist.

Each field contains atomic (indivisible) values.

Second Normal Form (2NF)

All non-key attributes depend fully on the primary key.

There are no partial dependencies.

Example: In the Players table, all attributes depend on PlayerID.

Third Normal Form (3NF)

No transitive dependencies exist.

Non-key attributes depend only on the primary key.

Example:

- City details are stored separately from countries.
- Stadiums reference cities instead of repeating city/country data.

This structure reduces redundancy and improves data consistency.

Physical Design

The database is implemented using Microsoft SQL Server as the database management system. The physical design focuses on efficient data storage, integrity, and performance.

Data is organized into relational tables with primary keys and foreign keys to enforce relationships between entities. Identity columns are used to automatically generate unique identifiers for records.

To ensure performance:

- Primary keys are automatically indexed
- Foreign keys improve join efficiency
- Queries are optimized using grouping, aggregation, and filtering techniques

Security is implemented through the creation of a login and database user, with controlled access permissions granted only to specific tables.

Data integrity is enforced through:

- Constraints (Primary Key, Foreign Key, NOT NULL, UNIQUE)
- Triggers (to prevent invalid or duplicate data)
- Transactions (to ensure safe updates)

Data Dictionary

Countries Table:

Column Name	Data Type	Description
CountryID	INT (PK, Identity)	Unique identifier for each country
CountryName	VARCHAR(100)	Name of the country

Cities Table:

Column Name	Data Type	Description
CityID	INT (PK, Identity)	Unique identifier for each city
CityName	VARCHAR(100)	Name of the city
CountryID	INT (FK)	References Countries table

Stadiums Table:

Column Name	Data Type	Description
StadiumID	INT (PK, Identity)	Unique stadium identifier
StadiumName	VARCHAR(100)	Name of stadium
Capacity	INT	Seating capacity
CityID	INT (FK)	References Cities table

Teams Table:

Column Name	Data Type	Description
TeamID	INT (PK, Identity)	Unique team identifier
TeamName	VARCHAR(100)	Name of team
CountryID	INT (FK)	References Countries
GroupName	CHAR(1)	Group (A–F)

Players Table:

Column Name	Data Type	Description
PlayerID	INT (PK, Identity)	Unique player identifier
PlayerName	VARCHAR(100)	Player name
Position	VARCHAR(50)	Playing position
TeamID	INT (FK)	References Teams

Matches Table:

Column Name	Data Type	Description
MatchID	INT (PK, Identity)	Unique match identifier

Column Name	Data Type	Description
MatchDate	DATE	Date of match
MatchTime	TIME	Time of match
Stage	VARCHAR(50)	Tournament stage
HomeTeamID	INT (FK)	Home team
AwayTeamID	INT (FK)	Away team
HomeScore	INT	Goals scored by home team
AwayScore	INT	Goals scored by away team
StadiumID	INT (FK)	Match venue

Tickets Table:

Column Name	Data Type	Description
TicketID	INT (PK, Identity)	Unique ticket identifier
MatchID	INT (FK)	References Matches
FanName	VARCHAR(100)	Name of fan
SeatNumber	VARCHAR(20)	Seat allocation
PurchaseDate	DATETIME	Date of purchase

Conclusion

The database is well-structured and follows proper relational database design principles.

It ensures:

- Data consistency through normalization
- Data integrity through constraints and triggers
- Efficient querying through relationships and indexing

The design accurately models a real-world FIFA World Cup tournament system and supports advanced database features such as views, stored procedures, functions, triggers, transactions, and security.